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NeuroCAPs: A Python Package for Performing Co-Activation Patterns Analyses on Resting-State and Task-Based fMRI Data

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Software

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Summary

Co-Activation Patterns (CAPs) is a dynamic functional connectivity technique that clusters similar spatial distributions of brain activity. To make this analytical technique more accessible to neuroimaging researchers, NeuroCAPs (D. Smith, 2025), an open source Python package, was developed. This package performs end-to-end CAPs analyses on preprocessed resting-state or task-based functional magnetic resonance imaging (fMRI) data, and is most optimized for data preprocessed with fMRIPrep, a robust preprocessing pipeline designed to minimize manual user input and enhance reproducibility (Esteban et al., 2019).

Background

Numerous fMRI studies employ static functional connectivity (sFC) techniques to analyze correlative activity within and between brain regions. However, these approaches operate under the assumption that functional connectivity patterns, which change within seconds (Jiang et al., 2022), remain stationary throughout the entire data acquisition period (Hutchison et al., 2013).

Unlike sFC approaches, dynamic functional connectivity (dFC) methods enable the analysis of dynamic functional states, which are characterized by consistent, replicable, and distinct periods of time-varying brain connectivity patterns (Rabany et al., 2019). Among these techniques, CAPs analysis aggregates similar spatial distributions of brain activity using clustering techniques, such as the k-means algorithm, to capture the dynamic nature of brain activity (Liu et al., 2018).

Statement of Need

Currently, performing an end-to-end CAPs analysis presents challenges due to the numerous steps required. Researchers must:

1. clean timeseries data through nuisance regression and censor frames with high framewise displacement (excessive head motion).
2. perform spatial dimensionality reduction.
3. concatenate timeseries data across multiple subjects for k-means clustering.
4. select an optimal cluster size using heuristics such as the elbow or silhouette methods.
5. implement various visualization techniques to enhance interpretability of the results.

While other excellent CAPs toolboxes exist, they are often implemented in proprietary languages such as MATLAB (which is the case for TbCAPs (Bolton et al., 2020)), lack comprehensive end-to-end analytical pipelines for both resting-state and task-based fMRI data with temporal

dynamic metrics and visualization capabilities (such as capcalc (Frederick & Drucker, 2022)), or are comprehensive, but generalized toolboxes for evaluating and comparing different dFC methods (such as pydFC (Torabi et al., 2024)). NeuroCAPs addresses these limitations by providing an accessible Python package specifically for performing end-to-end CAPs analyses, from post-processing of fMRI data to creation of temporal metrics for downstream statistical analyses and visualizations to facilitate interpretations. However, many of NeuroCAPs' post-processing functionalities assumes that fMRI data is organized in a Brain Imaging Data Structure (BIDS) compliant directory and is most optimized for data preprocessed with fMRIPrep (Esteban et al., 2019) or preprocessing pipelines that generate similar outputs (e.g. NiBabies (Goncalves et al., 2025)).

Modules

NeuroCAPs consists of four modules, with core functionality primarily distributed between two main modules (`neurocaps.extraction` and `neurocaps.analysis`) that handle the entire workflow, from post-processing to temporal metric computation and visualization capabilities.

neurocaps.exceptions

This module contains two custom exceptions: `BIDSQueryError` and `NoElbowDetected`. `BIDSQueryError` supports NeuroCAPs' integration with PyBIDS (Yarkoni et al., 2019), a tool for querying BIDs-compliant directories, providing user support guidance when subject IDs cannot be detected in BIDS directories. `NoElbowDetected` offers solutions when the elbow method (implemented via `KneeLocator` from Kneed (Arvai, 2023)) fails to identify the optimal cluster size for k-means.

neurocaps.extraction

This module contains the `TimeseriesExtractor` class, which:

- leverages extracts Nilearn's (contributors, n.d.) `NiftiLabelsMasker` to perform nuisance regression on resting-state and task-based fMRI data and use lateralized brain parcellations (such as the Schaefer (Schaefer et al., 2018), Automated Anatomical Labeling (Tzourio-Mazoyer et al., 2002), and Human Connectome Project extended (Huang et al., 2022) parcellations) for spatial dimensionality reduction.
- censors high-motion volumes using fMRIPrep-derived framewise displacement values and stores the extracted timeseries information in a dictionary mapping subject IDs to run IDs and their associated timeseries.
- reports the number of censored frames, interpolated frames, and mean and standard deviation of continuous high motion segments for each subject.
- saves extracted timeseries data as a pickle file.
- visualizes timeseries data for a specific subject's run.

neurocaps.analysis

This module contains the `CAP` class, which:

- allows group-specific analyses or analyses on all subjects.
- performs k-means clustering (from Scikit-learn (Pedregosa et al., 2011)) for CAP identification, while supporting a single cluster size or a range of clusters in combination with various cluster selection methods to determine the optimal cluster size.
- computes various subject-level temporal dynamics metrics for downstream statistical analyses.
- enables conversion of CAPs to NIfTI statistical maps.
- provides diverse visualization options, using Matplotlib (Hunter, 2007), Seaborn (Waskom, 2021), Plotly (Inc., n.d.), and Surfplot (Gale et al., 2021), to facilitate scientific interpretations.

Additionally, the module provides standalone functions for:

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- changing the data type (Harris et al., 2020) and performing additional standardization of timeseries data produced by `TimeseriesExtractor`.
- merging multiple timeseries data across different dictionaries produced by `TimeseriesExtractor` by identifying similar subjects and concatenating their data, which facilitates analyses to identify CAPs across sessions or tasks.
- creating averaged transition probability matrices from subject-level transition probabilities.

neurocaps.typing

This module provides custom type definitions compatible with static type checkers, enabling proper construction of dictionary structures for parcellations or timeseries data when manual creation is necessary.

Example Application

NeuroCAPs was originally developed (and later refined for broader use) to facilitate the analysis in D. D. Smith et al. (2025), which has been submitted for peer review by the same author. In this manuscript, NeuroCAPs was used to extract timeseries data, cluster and identify CAPs using the elbow method, compute temporal metrics for downstream statistical analyses, and produce visualizations for CAPs (i.e., heatmap, surface plots, correlation matrix, and radar plots).

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